

amount for each of these outcomes, then the expected value could be defined as the probability of a win times the average win size minus the probability of a loss times the average loss size. For example, if you estimate the probability of a win at 20 percent, and a winning outcome is expected to be \$10,000 and a losing outcome \$1,000, then the expected gain for the trade is +\$1,200 (derived as follows:  $0.2 * \$10,000 - 0.8 * \$1,000 = \$1,200$ ).

<sup>7</sup>CDOs were a general type of securitization that were also built from many other types of instruments, but these other constructions are not germane to this discussion.

<sup>8</sup>Although correlations for individual mortgages could also be significant during severe economic downturns, the degree of correlation would still not be nearly as extreme as the correlations between different BBB tranches.

<sup>9</sup>Mortgage-backed securitizations (MBS) are a subset of asset-backed securitizations (ABS).

<sup>10</sup>As explained above, the CDO combined many BBB tranches and created AAA and AA tranches consisting of nothing but these low-rated MBS tranches. The justification for this transformation was that the BBB tranches of different securitizations were uncorrelated. This assumption, however, made no sense at all because if the BBB tranche of one subprime mortgage securitization became impaired, it was highly likely that the BBB tranches of other subprime securitizations would also become impaired. The higher rating of the top CDO tranches meant that CDS protection could be bought far cheaper, making the CDOs a more attractive vehicle than MBS for establishing short subprime mortgage positions.

<sup>11</sup>The higher ratings assigned to the upper-tier tranches of CDOs implied (incorrectly) that these bonds had a low chance of default. Therefore, Cornwall could buy protection that paid off if these bonds defaulted for only the small premium associated with bonds with AAA and AA ratings. Since the premium paid for the default protection is the maximum risk in the short position, the CDOs provided a way to short the worst quality subprime bonds with minimal risk exposure.